RAHUL MITRA

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EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Software Engineering, Data Structures and Algorithms, Artificial Intelligence, 3D Computer Vision, Operations Research, Simulation, Statistics, Linear Algebra.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine ingesting each site's shelf coordinates, dimensions, and operating rules, letting one model adapt to any STMicroelectronics warehouse floor plan without per-site rework.

Global Data Engineer (Operational AI Systems), Part-Time, Abbott Laboratories (Contract) **Jun 2026 – Present**

- Delivered AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.
- Strengthened monitoring and process structure across healthcare supply chains with AI, protecting product availability for vulnerable patient groups.

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Deployed a team-built APC simulator to production for live internal users on Azure App Service, with Docker packaging, access restrictions, and runtime and session hardening.

Software Engineer (Full Stack & Cloud), People's Association (Pro bono) **Sep 2025 – Sep 2026**

- Recruited into PA's Sparks Citizens Developer network as a community innovator in recognition of prior public-good initiatives, then designed, built, and deployed Churp, a resident-driven garden-plot balloting platform now live in Tengah and slated for national rollout, seeded by a S$20K Sparks Community Innovation Fund award.
- Integrated Singpass Login (OIDC) to secure resident sign-in and Myinfo to verify identity and address, keeping garden-plot balloting fair for 150 residents signed up at release.
- Surveyed garden sites into geospatial datasets for mapping, plot allocation, and site planning, and designed rules-based balloting across 4 control points (eligibility, conflict resolution, exceptions, admin oversight), keeping allocation auditable.
- Provisioned 10 AWS services with Terraform (ECS Fargate, Multi-AZ RDS Postgres with PostGIS, ALB behind WAFv2, Route 53, ACM, KMS, Secrets Manager, and a Rust auth gateway on Lambda and API Gateway), shipped via keyless GitHub Actions CI/CD on IAM OIDC across 5 repos with gated migrations, RDS snapshots, and auto-rollback.

RESEARCH EXPERIENCE

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.

PROJECTS

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

SKILLS AND CERTIFICATIONS

Technical: Solution architecture; cloud (AWS Fargate, Terraform, Azure, Docker, CI/CD); full-stack (React, TypeScript, FastAPI, Flask); Singpass Login & Myinfo (OIDC); RAG & AI agents; AI tools (Claude Code, ChatGPT, Codex, Gemini, ComfyUI); GIS mapping; accessibility; rapid prototyping
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery